

- **Accuracy:** How close a measured value is to the **true or accepted value**.
 - Think: "Did I hit the target?"
- **Precision:** How close **multiple measurements** are to each other, even if they are not close to the true value.
 - Think: "Are my results consistent?"

Accuracy

Accuracy means how close a measurement is to the **true or correct value**.

Example:

The actual length of a pencil is **15 cm**.

Measurement	Accurate?
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15 cm	Yes
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14.9 cm	Very close
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12 cm	No
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Accuracy = Correctness

Think: **Did I hit the target?**

Precision

Precision means how close repeated measurements are to **each other**.

Example:

Three students measure the same pencil.

Measurement: 14.2 cm, 14.2 cm, 14.2 cm

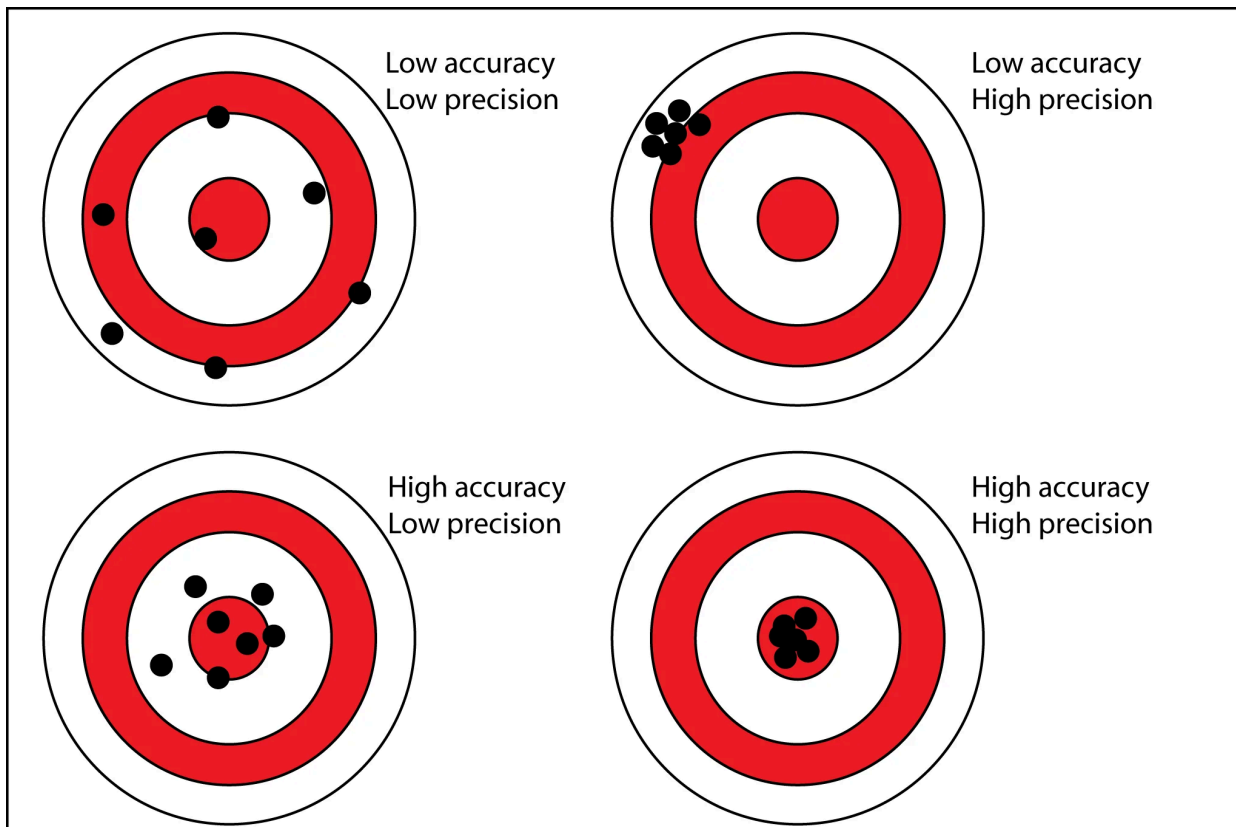
All measurements are the same. But it is precise as true length is 15 cm so they are not accurate

Precision=Consistency

Did I get the same result every time

When you throw darts at a dartboard:

- **High accuracy + high precision:** All darts are close together *and* near the bullseye.
- **High precision + low accuracy:** All darts are close together, but far from the bullseye.
- **Low precision + high accuracy:** Darts are spread out but average near the bullseye.
- **Low accuracy + low precision:** Darts are scattered and far from the bullseye.



Real-life Classroom Example:

Measuring the length of a table 3 times:

- Results: 150.2 cm, 150.3 cm, 150.2 cm
(Precision = High)
- Actual length = 152.0 cm
(Accuracy = Low)

Task 1: Definitions

Fill in the blanks using the word bank:

Accuracy, Precision, Target, Measurements, Same

1. _____ is how close a result is to the correct value.
2. _____ is when repeated measurements are very close to each other.
3. A dart hitting the center of a dartboard shows good _____.
4. If you take three measurements and they are all the _____, you show good _____.

Task 2: Analyze and Reflect

Your classmate measured the mass of a fruit 3 times:

- Trial 1: 152g
 - Trial 2: 151.9g
 - Trial 3: 152.1g
- The actual value was 160g.

Answer the following:

1. Are the results accurate? Why or why not?

2. Are the results precise? Why or why not?

3. What could be done to improve **accuracy** in this case?

Task: 3 Suppose the actual mass of an apple is **200 g**.

Students	Values obtained	Accurate/Precise/ High/Low
Student A	200 g, 201 g, 199 g	
Student B	180 g, 180 g, 180 g	
Student C	190 g, 210 g, 200 g	